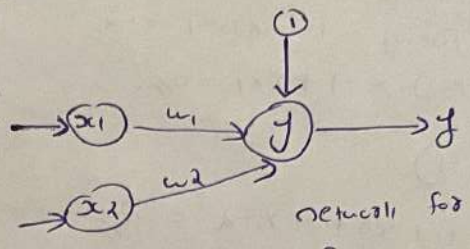
3a) Find the weights using the perception network for AND NOT function when all the i/ps are presented only once use bipolar i/ps & targets

x_1	x_2	t
1	1	-1
1	-1	1
-1	1	-1
-1	-1	-1



network for AND NOT function

Let $x=1$ $\theta=0$

(1, 1, -1)

$y_{in} = b + x_1w_1 + x_2w_2 = 0 + 1\times0 + 1\times0 = 0$

Applying activation function over the net i/p

$$y = f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} > 0 \\ 0 & \text{if } -0 \le y_{in} \le 0 \\ -1 & \text{if } y_{in} < -0 \end{cases}$$

$w_1(new) = w_1 old + \alpha t x_1 = 0 + 1 \times -1 \times 1 = -1$

$w_2(new) = -1$

$b(new) = b old + \alpha t = 0 + 1 \times -1 = -1$

$w = [-1, -1, -1]$

(1, -1, 1)

$y_{in} = b + x_1w_1 + x_2w_2$

$= -1 - 1 + 1 = -1$

(-1, 1, 1)

$w_1(new) = -1 + 1 \times 1 \times 1 = 0$

$w_2(new) = 1 \times 1 \times 1 \times -1 = -2$

$b(new) = -1 + 1 \times 1 = 0$

(-1, -1, -1)

$y_{in} = 0 + -1 \times 0 + -1 \times -2 = 2$

$y = f(y_{in}) = 1$

$w_1(new) = w_1(old) + \alpha t x_1 = 0 + 1 \times -1 \times -1 = 1$

$w_2(new) = w_2(old) + \alpha t x_2 = -2 + 1 \times -1 \times -1 = -1$

weights after presenting fourth i/p sample are

$w = [1, -1, 1]$

one epoch of training for AND NOT function

i/p			Target (t)	Net i/p y_{in}	Calculated output (y)	weights w_1 (0)	w_2 (0)	b (0)
x_1	x_2	b						
1	1	1	-1	0	0	-1	-1	-1
1	-1	1	1	-1	-1	0	-2	0
-1	1	1	-1	-2	-1	0	-2	0
-1	-1	1	-1	2	1	1	-1	-1

8b) How is training algorithm performed in back propagation neural network.

Step 0: initialize weights and learning rate

Step 1: Perform step 2-9 when stopping condition is false

Step 2: Perform step 3-8 for each training pairs

Step 3 : Each i/p unit receives i/p signal and sends to the hidden unit

Step 4 : each hidden unit z_j ($j=1$ to p) sums its weighted i/p signals to calculate net i/p

$$z_{in_j} = v_{0j} + \sum_{i=1} x_i v_{ij}$$

Step 5 :- For each output unit y_k ($k=1$ to m)

Calculate the net i/p

$y_{in_k} = w_{0k} + \sum_{j=1} z_j w_{jk}$ and apply the activation function to compute the output signal

$y_k = f(y_{in_k})$

Back propagation of error (phase II) :

Step 6 : each output unit y_k ($k=1$ to m) receives a target pattern corresponding to i/p training

$$\delta_k = (t_k - y_k) f'(y_{in_k})$$

Update the weights

$$\Delta w_{jk} = \alpha \delta_k z_j$$

$$\Delta w_{0k} = \alpha \delta_k$$

Step 7 : each hidden unit ($j=1$ to p) sums its inputs from the output units

Then update the weights and bias

$$\Delta v_{ij} = \alpha \delta_j x_i \qquad \Delta v_{0j} = \alpha \delta_j$$

weight & bias updation (phase II)

Step 8 : each output unit y_k ($k=1$ to m) updates the bias and weights

$$v_{ij}(new) = v_{ij}(old) + \Delta v_{ij}$$

$$v_{0j}(new) = v_{0j}(old) + \Delta v_{0j}$$

Step 9: check for the stopping condition may be certain number of epochs reached or when the actual output equals to target output